

LeadHunter AI

Autonomous B2B Lead Engine & Local Outreach Platform — Operations Manual

1. Executive Overview

LeadHunter AI is an autonomous agent designed for digital agencies, web design freelancers, and growth consultants. The system scans local businesses via Google Maps (SerpAPI), classifies online presence (missing, broken, or social-only websites), scores prospect urgency, builds live mobile demo sites on custom preview URLs, and prepares cold outreach via Gmail SMTP and WhatsApp.

2. Pipeline Architecture (11 Stages)

Stage	Name	Key Functionality
1	Discovery	Queries SerpAPI Google Maps; extracts phone, address, rating, review count.
2	Dedup & Normalization	Strips +91, cleans URLs, generates deterministic SHA256 lead_id hash.
3	Website Verification	HTTP crawler detects NO_WEBSITE, BROKEN_WEBSITE, SOCIAL_ONLY, VALID_WEBSITE.
4	Scoring & Tiering	Assigns 0-100 score: HOT (≥70), WARM (45-69), LOW (<45).
5	AI Personalization	Generates custom subject lines, cold email, & WhatsApp copy via Claude/Template.
6	Demo Website Engine	Creates single-page mobile preview websites hosted on /preview/{slug}.
7	Google Sheets Sync	Synchronizes 26 columns to 'Leads' tab and batch logs to 'Runs' tab.
8	Human Approval Gate	Interactive Web UI & terminal cards ([A]pprove / [R]eject / [S]kip).
9	Outbound Outreach	Dispatches emails (Gmail SMTP) & WhatsApp with rolling hourly rate limits.
10	Follow-Up Lifecycle	Automated Day 3 (Follow-up 1), Day 7 (Follow-up 2), and Day 10 (Cold) schedule.

3. Prospect Scoring Engine

Signal / Condition	Points Awarded	Qualification Tier
NO_WEBSITE (No site found)	+40 pts	HOT (Score ≥ 70) — Immediate Outreach
BROKEN_WEBSITE (HTTP 4xx/5xx/SSL)	+35 pts	HOT (Score ≥ 70) — High Urgency Pitch
SOCIAL_ONLY (Facebook / IG only)	+30 pts	WARM (Score 45-69) — Opportunity Pitch
Has Phone Number (10 digits)	+15 pts	Required for WhatsApp Dispatch
Rating 4.0+ with 20+ Reviews	+15 pts	Strong local reputation signal
Target Niche (Restaurant, Clinic, Salon, Gym)	+10 pts	High web-conversion sector
VALID_WEBSITE (Active custom website)	0 pts	LOW (< 45 pts) — Archived in Database

4. Key Operational Commands

1. Start Web Control Center (Port 8500):

```
python main.py --web --port 8500
```

2. Start Cloudflare Public Tunnel (Allows mobile preview):

```
cloudflared tunnel --url http://localhost:8500
```

3. Run Full Pipeline for Any City (CLI Mode):

```
python main.py --all --city "Chandigarh" --category "restaurants" --limit 5
```

4. Run Automated Follow-Up Cycle:

```
python main.py --followups
```

5. Environment Configuration (.env Reference)

```
# --- Google Maps Discovery ---
SERPAPI_KEY=8b9e65a6e530034833b3d1f0b555227a54e956330dc6de4324ca95a182efd643

# --- Public Demo Base URL (Cloudflare Tunnel) ---
DEMO_BASE_URL=https://your-tunnel.trycloudflare.com

# --- Email Outreach (Gmail SMTP) ---
SENDER_EMAIL=your-real-email@gmail.com
SENDER_NAME=Your Agency Name
GMAIL_APP_PASSWORD=xxxx xxxx xxxx xxxx # 16-character App Password

# --- Safety Guards ---
DRY_RUN=true # Set false only when ready for real dispatch
```

6. Safety & Anti-Spam Compliance Guardrails

- **DRY_RUN Safety Net:** Outreach execution is simulated by default unless `DRY_RUN=false` is explicitly set.
- **Mandatory Human Approval Gate:** System strictly prevents email/WhatsApp dispatch unless `approval_status` is `APPROVED`.
- **Rate Limiting:** Cap of 10 emails/hour and 20 WhatsApp messages/hour with 3-second inter-message delay.
- **Opt-out & Reply Tracking:** Leads marked as `REPLIED` or `DO_NOT_CONTACT` are permanently excluded from follow-ups.